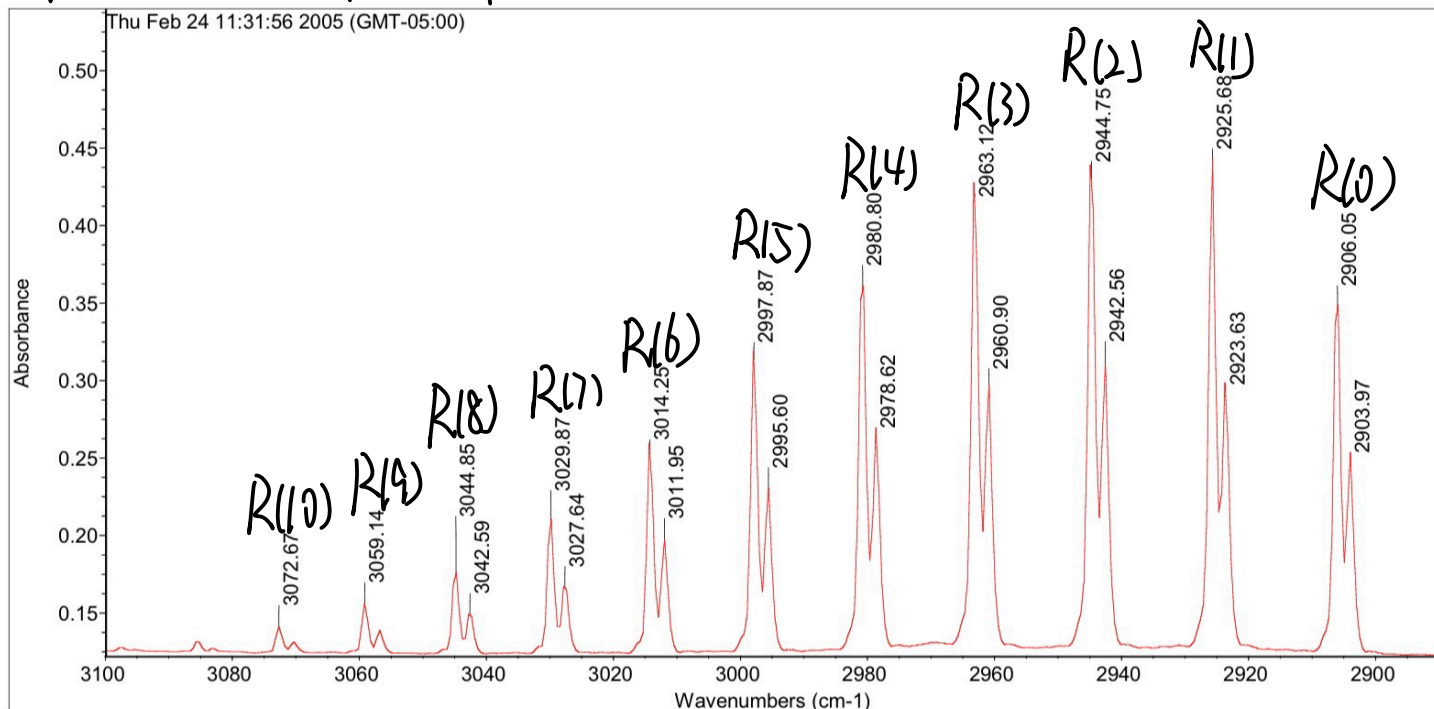


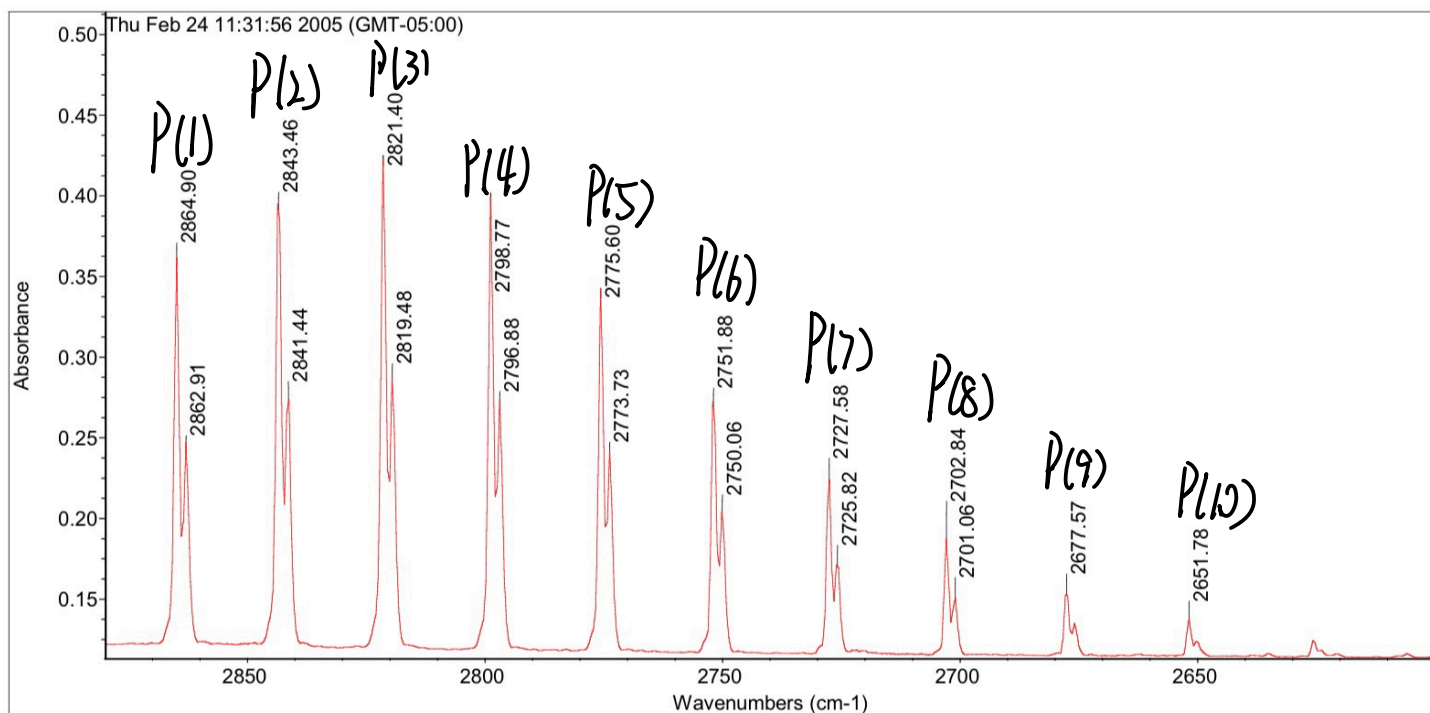
Lab 4 Vibrational-Rotational Spectra of Diatomic Molecules

1. Calculations for HCl

* R branch of HCl



* P Branch of HCl



Assume intercept as b , slope as m , i.e. $y = mx + b$

Using the data above for regression, from Eq. 25:

$$B_1 = b = 10.13496 \pm 0.00051 \text{ cm}^{-1}$$

$$D_e^{25} = -\frac{1}{2}m = (8.15000 \pm 0.05000) \times 10^{-4} \text{ cm}^{-1}$$

From Eq. 26:

$$B_0 = b = 10.43743 \pm 0.00117 \text{ cm}^{-1}$$

$$D_e^{26} = -\frac{m}{2} = (8.10000 \pm 0.10000) \times 10^{-4} \text{ cm}^{-1}$$

From Eq. 27:

$$\tilde{\nu}_{B0} = b = (2885.78919 \pm 0.00567) \text{ cm}^{-1}$$

$$B_1 - B_0 = m = (-0.30341 \pm 0.00011) \text{ cm}^{-1}$$

From Eq. 9:

$$B_e = b = (1058441 \pm 0.00225) \text{ cm}^{-1}$$

$$\alpha_e = -m = (0.29925 \pm 0.00065) \text{ cm}^{-1}$$

From Eq. 30:

$$\tilde{\nu}_0 = (2988.95306 \pm 0.17451) \text{ cm}^{-1}$$

$$\tilde{\nu}_0 \alpha_e = (51.63915 \pm 0.04113) \text{ cm}^{-1}$$

$$\text{As } \tilde{\nu}_0 = \tilde{\nu}_e - 2\tilde{\nu}_e \alpha_e, \tilde{\nu}_0 \alpha_e = \tilde{\nu}_e \alpha_e -$$

$$\tilde{\nu}_e = \tilde{\nu}_0 + 2\tilde{\nu}_0 \alpha_e$$

$$\Rightarrow \tilde{\nu}_e = 2988.95306 + 2 \cdot 51.63915 = 3092.23136 \text{ cm}^{-1}$$

$$S_{\tilde{\nu}_e} = \sqrt{S_{\tilde{\nu}_0}^2 + 4S_{\tilde{\nu}_0 \alpha_e}^2} = \sqrt{(0.17451)^2 + 4 \cdot (0.04113)^2} \approx 0.19293 \text{ cm}^{-1}$$

$$S_0, \tilde{\nu}_e = (3092.23136 \pm 0.19293) \text{ cm}^{-1}$$

$$\text{Reduced mass: } \mu = \frac{m_H \cdot m_{Cl}}{m_H + m_{Cl}} = \frac{1.00794 \cdot 35.453}{1.00794 + 35.453} \text{ a.u.} \approx 1.62665 \times 10^{-27} \text{ kg}$$

$$\text{From } B_e = \frac{h}{8\pi^2 c \mu r_e^2}, \quad r_e = \sqrt{\frac{h}{8\pi^2 c \mu B_e}} = \sqrt{\frac{6.62607 \times 10^{-34}}{8\pi^2 \cdot 2.99792 \times 10^8 \cdot 1.62665 \times 10^{-27} \cdot 10.5844}} \text{ m}$$

$$\approx 1.27509 \times 10^{-10} \text{ m} = 1.27509 \text{ \AA}$$

$$S_{r_e} = \frac{1}{2} r_e \cdot \frac{S_{B_e}}{B_e} = 1.27509 \cdot \frac{1}{2} \cdot \frac{0.02225}{10.5844} = 0.00014 \text{ \AA}$$

$$S_0, r_e = 1.27509 \pm 0.00014 \text{ \AA}$$

$$\text{From } \tilde{\nu}_e = \frac{1}{2\pi c} \sqrt{\frac{k}{\mu}}, \quad k = (2\pi c \tilde{\nu}_e)^2 \mu = (2\pi \cdot 2.99792 \times 10^8 \cdot 3.09223 \times 10^5)^2 \cdot 1.62665 \times 10^{-27} \text{ N}\cdot\text{m}^{-1}$$

$$\approx 55187366 \text{ N}\cdot\text{m}^{-1}$$

$$S_k = 2k \frac{S_{r_e}}{r_e} = 2 \cdot 55187366 \cdot \frac{0.00014}{1.27509} \approx 206886 \text{ N}\cdot\text{m}^{-1}$$

$$S_0, k = (55187366 \pm 206886) \text{ N}\cdot\text{m}^{-1}$$

$$\text{Averaged } D_e: \bar{D} = \frac{1}{2} (D_e^{25} + D_e^{26}) = \frac{1}{2} \cdot (5.15000 + 5.10000) \times 10^4 \text{ cm}^{-1}$$

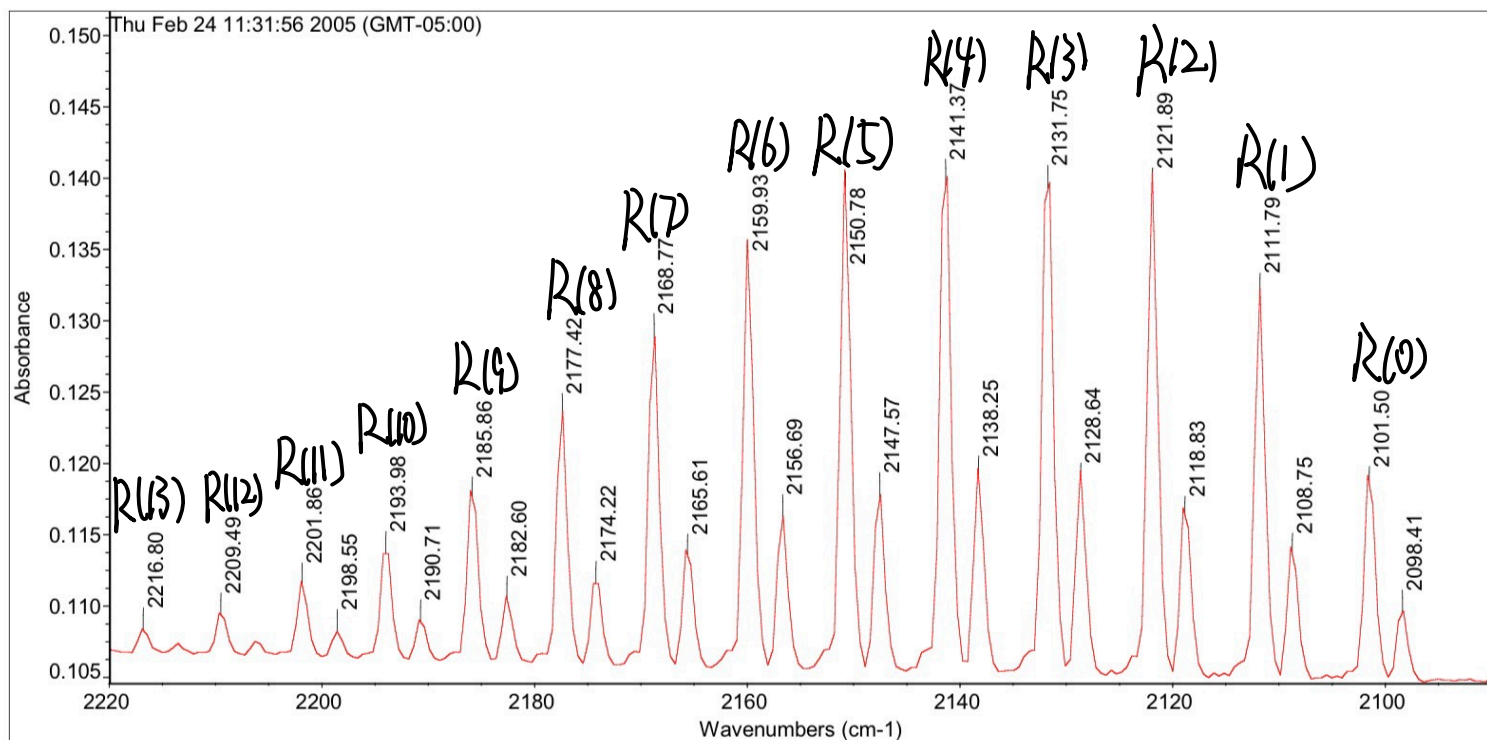
$$= 5.12500 \times 10^4 \text{ cm}^{-1}$$

$$S_{\bar{D}} = \frac{1}{2} \sqrt{S_{D_e^{25}}^2 + S_{D_e^{26}}^2} = \frac{1}{2} \sqrt{0.05^2 + 0.1^2} \times 10^4 \approx 0.05600 \times 10^4$$

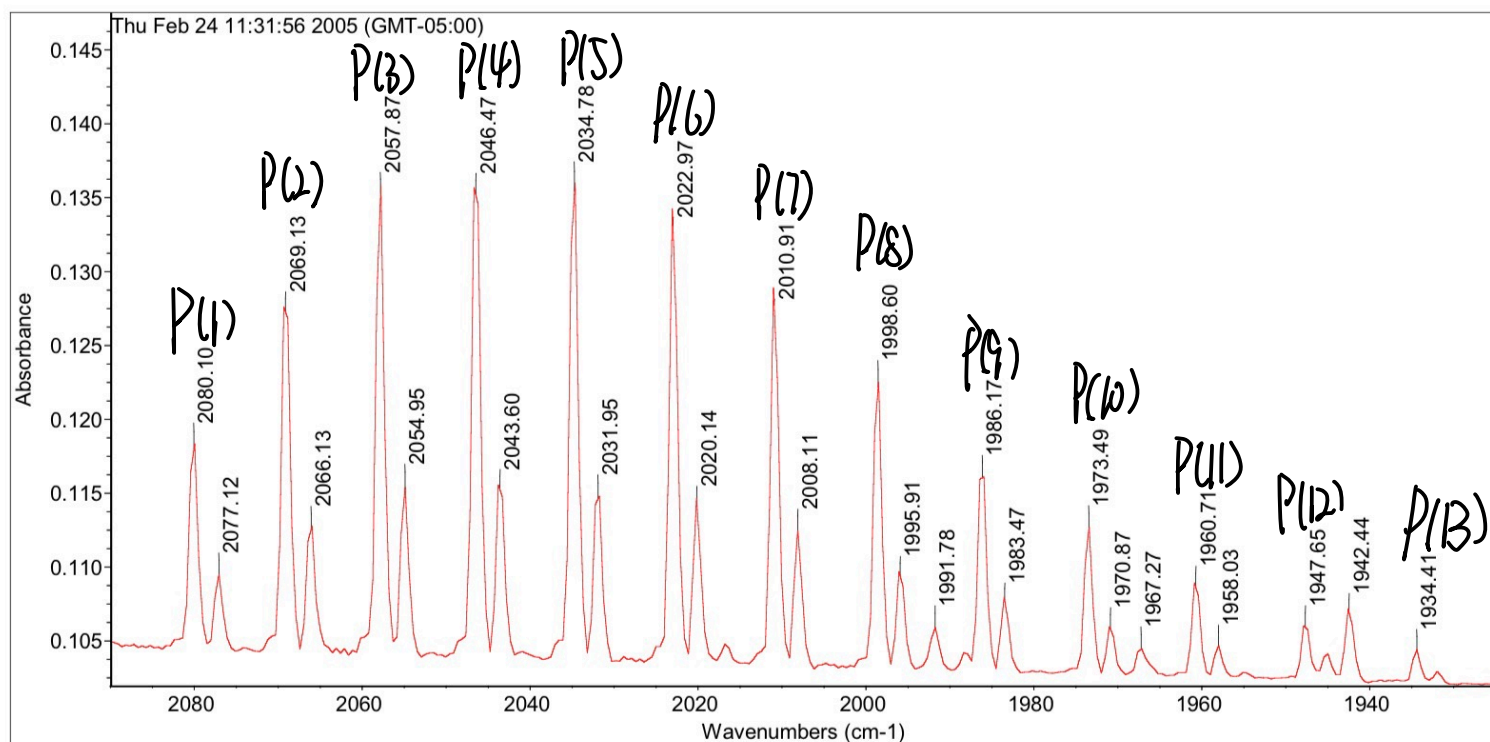
$$S_0, D_e = (5.12500 \pm 0.05600) \times 10^4 \text{ cm}^{-1}$$

2. Calculation for DCI

*R branch of DCI



*P branch of DCI



Using the data above for regression, from Eq. 25:

$$B_1 = b = 5.27980 \pm 0.00063 \text{ cm}^{-1}$$

$$D_e^{25} = -\frac{m}{2} = (1.40000 \pm 0.05000) \times 10^{-4} \text{ cm}^{-1}$$

From Eq. 26:

$$B_0 = b = 5.39285 \pm 0.00072 \text{ cm}^{-1}$$

$$D_e^{26} = -\frac{m}{2} = (1.45000 \pm 0.05000) \times 10^{-4} \text{ cm}^{-1}$$

From Eq. 27:

$$\tilde{\nu}_{B0} = b = (2090.89783 \pm 0.00744) \text{ cm}^{-1}$$

$$B_1 - B_0 = m = (-0.1124 \pm 0.00009) \text{ cm}^{-1}$$

From Eq. 9:

$$B_e = b = (5.44881 \pm 0.00067) \text{ cm}^{-1}$$

$$\alpha_e = -m = (0.11237 \pm 0.00039) \text{ cm}^{-1}$$

From Eq. 30:

$$\tilde{\nu}_0 = (2144.18906 \pm 0.05555) \text{ cm}^{-1}$$

$$\tilde{\nu}_0 \alpha_e = (26.65067 \pm 0.01780) \text{ cm}^{-1}$$

As $\tilde{\nu}_0 = \tilde{\nu}_e - 2\tilde{\nu}_e \alpha_e$, $\tilde{\nu}_0 \alpha_e = \tilde{\nu}_e \alpha_e -$

$$\tilde{\nu}_e = \tilde{\nu}_0 + 2\tilde{\nu}_0 \alpha_e$$

$$\Rightarrow \tilde{\nu}_e = 2144.18906 + 2 \cdot 26.65067 = 2197.49040 \text{ cm}^{-1}$$

$$S_{\tilde{\nu}_e} = \sqrt{S_{\tilde{\nu}_0}^2 + 4S_{\tilde{\nu}_0 \alpha_e}^2} = \sqrt{(0.05555)^2 + 4 \cdot (0.01780)^2} \approx 0.06581 \text{ cm}^{-1}$$

$$S_0, \tilde{\nu}_e = (2197.49040 \pm 0.06581) \text{ cm}^{-1}$$

$$\text{Reduced mass: } \mu = \frac{m_H \cdot m_{Cl}}{m_H + m_{Cl}} = \frac{2.01410 \cdot 35.453}{2.01410 + 35.453} \text{ a.u.} \approx 3.16235 \times 10^{-27} \text{ kg}$$

$$\text{From } B_e = \frac{h}{8\pi^2 c \mu r_e^2}, \quad r_e = \sqrt{\frac{h}{8\pi^2 c \mu B_e}} = \sqrt{\frac{6.62607 \times 10^{-34}}{8\pi^2 \cdot 2.99792 \times 10^8 \cdot 3.16235 \cdot 10^{-27} \cdot 544.881}} \text{ m}$$

$$\approx 1.27458 \times 10^{-10} \text{ m} = 1.27458 \text{ \AA}$$

$$S_{r_e} = \frac{1}{2} r_e \cdot \frac{S_{B_e}}{B_e} = 1.27458 \cdot \frac{1}{2} \cdot \frac{0.00067}{5.44881} = 0.00008 \text{ \AA}$$

$$S_0, r_e = 1.27458 \pm 0.00008 \text{ \AA}$$

$$\text{From } \tilde{\nu}_e = \frac{1}{2\pi c} \sqrt{\frac{k}{\mu}}, \quad k = (2\pi c \tilde{\nu}_e)^2 \mu = (2\pi \cdot 2.99792 \times 10^8 \cdot 2.19749 \times 10^5)^2 \cdot 3.16235 \times 10^{-27} \text{ N}\cdot\text{m}^{-1}$$

$$\approx 541.83065 \text{ N}\cdot\text{m}^{-1}$$

$$S_k = 2k \frac{S_{\tilde{\nu}_e}}{\tilde{\nu}_e} = 2 \cdot 541.83065 \cdot \frac{0.06581}{2197.49040} \approx 0.03245 \text{ N}\cdot\text{m}^{-1}$$

$$S_0, k = (541.83065 \pm 0.03245) \text{ N}\cdot\text{m}^{-1}$$

$$\text{Averaged } D_e: \bar{D} = \frac{1}{2} (D_e^{25} + D_e^{26}) = \frac{1}{2} \cdot (1.40000 + 1.45000) \times 10^{-4} \text{ cm}^{-1}$$

$$= 1.42500 \times 10^{-4} \text{ cm}^{-1}$$

$$S_{\bar{D}} = \frac{1}{2} \sqrt{S_{D_e^{25}}^2 + S_{D_e^{26}}^2} = \frac{1}{2} \sqrt{0.0570205 \times 10^{-4}} \approx 0.03540 \times 10^{-4}$$

$$S_0, D_e = (1.42500 \pm 0.03540) \times 10^{-4} \text{ cm}^{-1}$$

3. Calculations for Error Rate

(1) HC1

$$① B_1^{lit} = 10.136223$$

$$\text{error rate} = \frac{|10.13466 - 10.136223|}{10.136223} \times 100\% \approx 0.1246\%$$

$$② B_0^{lit} = 10.440254$$

$$\text{error rate} = \frac{|10.43743 - 10.440254|}{10.440254} \times 100\% \approx 0.2705\%$$

$$③ \tilde{U}_0(40)^{lit} = 2883.97250$$

$$\text{error rate} = \frac{|2883.7891 - 2883.97250|}{2883.97250} \times 100\% \approx 0.00652\%$$

$$④ B_e^{lit} = 10.593404$$

$$\text{error rate} = \frac{|10.58441 - 10.593404|}{10.593404} \times 100\% \approx 0.08490\%$$

$$⑤ \alpha_e^{lit} = 0.307139$$

$$\text{error rate} = \frac{|0.29925 - 0.307139|}{0.307139} \times 100\% \approx 2.5654\%$$

$$⑥ B^{lit} = 2990.97424$$

$$\text{error rate} = \frac{|2988.95306 - 2990.97424|}{2990.97424} \times 100\% \approx 0.6758\%$$

$$⑦ (\tilde{V}_e \tilde{\alpha}_e)^{lit} = 52.84579$$

$$\text{error rate} = \frac{|51.63915 - 52.84579|}{52.84579} \times 100\% \approx 2.2832\%$$

$$⑧ D_e^{lit} = 5.32019$$

$$\text{error rate} = \frac{|5.12500 - 5.32019|}{5.32019} \times 100\% \approx 3.6685\%$$

$$① r_e^{\text{lit}} = 1.27460 \text{ \AA}$$

$$\text{error rate} = \frac{|1.27509 - 1.27460|}{1.27460} \times 100\% \approx 0.03844\%$$

$$② k^{\text{lit}} = 516 \text{ N}\cdot\text{m}^{-1}$$

$$\text{error rate} = \frac{|515.87366 - 516|}{516} \times 100\% \approx 6.95226\%$$

② DCI

$$① B_1^{\text{lit}} = 5.279816$$

$$\text{error rate} = \frac{|5.27980 - 5.279816|}{5.279816} \times 100\% \approx 0.00030\%$$

$$② B_0^{\text{lit}} = 5.392261$$

$$\text{error rate} = \frac{|5.392285 - 5.392261|}{5.392261} \times 100\% \approx 0.01092\%$$

$$③ \tilde{\nu}_{\text{CO}}^{\text{lit}} = 2091.06130$$

$$\text{error rate} = \frac{|2090.89783 - 2091.06130|}{2091.06130} \times 100\% \approx 0.00782\%$$

$$④ B_e^{\text{lit}} = 5.448794$$

$$\text{error rate} = \frac{|5.44881 - 5.448794|}{5.448794} \times 100\% \approx 0.00029\%$$

$$⑤ \alpha_e^{\text{lit}} = 0.113291$$

$$\text{error rate} = \frac{|0.11237 - 0.113291|}{0.113291} \times 100\% \approx 0.81295\%$$

$$⑥ B^{\text{lit}} = 2145.16300$$

$$\text{error rate} = \frac{|2144.18906 - 2145.16300|}{2145.16300} \times 100\% \approx 0.04540\%$$

$$⑦ (\tilde{\nu}_{\text{e}}\tilde{\nu}_{\text{e}})^{\text{lit}} = 27.18252$$

$$\text{error rate} = \frac{|26.65067 - 27.18252|}{27.18252} \times 100\% \approx 1.9559\%$$

$$⑧ D_e^{\text{lit}} = 1.400$$

$$\text{error rate} = \frac{|1.42500 - 1.400|}{1.400} \times 100\% \approx 1.78571\%$$

$$⑨ r_e^{\text{lit}} = 1.274 \text{ \AA}$$

$$\text{error rate} = \frac{|1.27458 - 1.274|}{1.274} \times 100\% \approx 0.04553\%$$

$$⑩ k^{\text{lit}} = 516 \text{ N}\cdot\text{m}^{-1}$$

$$\text{error rate} = \frac{|541.8305 - 516|}{516} \times 100\% \approx 5.00594\%$$

4. Calculation for TCl properties

$$\text{TCl}_1 \mu = \frac{m_T m_{\text{Cl}}}{m_T + m_{\text{Cl}}} = \frac{32.05 \cdot 35.453}{32.05 + 35.453} \approx 17.7959 \text{ au}$$

$$\text{HCl}_1 \mu = \frac{m_H m_{\text{Cl}}}{m_H + m_{\text{Cl}}} = \frac{1.00794 \cdot 35.453}{1.00794 + 35.453} \approx 0.98008$$

$$\text{Since } \tilde{\nu} \propto \sqrt{\mu}, \quad \tilde{\nu}(\text{TCl}) = \tilde{\nu}(\text{HCl}) \sqrt{\frac{\mu_{\text{HCl}}}{\mu_{\text{TCl}}}}$$

$$\text{Use data from HCl, } \tilde{\nu}_e = 3092.23136 \text{ cm}^{-1}$$

$$\tilde{\nu}(\text{TCl}) = 3092.23136 \cdot \sqrt{\frac{0.98008}{17.7959}} \approx 1836.16783$$